

BLUESTOCK FINTECH

Mutual Fund Analytics

Capstone Project — Final Report

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1. Executive Summary

This report presents a comprehensive end-to-end mutual fund analytics project developed during the Bluestock Fintech internship. The project covers 40 mutual fund schemes with daily NAV data spanning 2019 to 2024, comprising 1,566 trading days of data analysis.

The primary objective was to build a complete analytics pipeline covering data ingestion, cleaning, exploratory analysis, performance metrics computation, risk analytics, investor behavior analysis, and interactive dashboard development.

Key Achievements:

Metric	Result
Funds Analyzed	40 schemes across Large Cap, Mid Cap, Small Cap & Flexi Cap
Data Period	January 2019 to December 2024 (6 years)
Metrics Computed	CAGR, Sharpe, Sortino, Alpha, Beta, VaR, CVaR, Max Drawdown
Top Fund Score	Tata Large Cap Fund — 88.88/100
Best Sharpe Ratio	Canara Robeco Bluechip — 1.50
Dashboard Pages	4 interactive Power BI pages
Investor Records	1,270 transactions across 93 investors

2. Data Sources & ETL Design

The project utilizes multiple data sources to build a comprehensive mutual fund analytics platform. All data was processed through a systematic ETL pipeline built in Python.

2.1 Data Sources

Source	Description	Records
nav_data.csv	Daily NAV for 40 mutual fund schemes	62,640 rows
nifty50.csv	Nifty 50 benchmark daily close prices	1,566 rows
nifty100.csv	Nifty 100 benchmark daily close prices	1,566 rows
expense_ratio.csv	Annual TER for all 40 fund schemes	40 rows
investor_data.csv	SIP and lumpsum transaction records	1,270 rows
sector_weights.csv	Sector allocation weights per fund	400 rows

2.2 ETL Pipeline Design

The ETL pipeline follows a structured approach: (1) Data Ingestion — raw CSV files loaded using Pandas with date parsing and index setting; (2) Data Cleaning — handling missing values, outlier detection, and data type validation; (3) Data Transformation — computing derived metrics including returns, ratios, and rankings; (4) Data Loading — exporting processed data to CSV files for dashboard consumption.

3. Exploratory Data Analysis

Comprehensive EDA was performed on all 40 fund schemes to understand return distributions, correlation patterns, and investor behavior.

3.1 Return Distribution Analysis

Daily returns for all 40 funds follow an approximately normal distribution centered near 0.04% per day, consistent with efficient market hypothesis. The distribution validates that our data is clean and suitable for statistical analysis.

3.2 CAGR Analysis

Fund Name	1yr CAGR	3yr CAGR	5yr CAGR
Tata Large Cap Fund	42.1%	85.3%	71.2%
DSP Midcap Fund	38.4%	75.5%	64.8%
ICICI Pru Midcap	35.9%	72.2%	61.3%
Nippon India Small Cap	31.2%	65.7%	58.9%
Canara Robeco Bluechip	29.8%	61.5%	55.4%

4. Fund Performance Analysis

4.1 Sharpe & Sortino Ratios

Sharpe ratio measures risk-adjusted return using total volatility, while Sortino ratio focuses only on downside risk. Risk-free rate used: 6.5% (RBI Repo Rate proxy), annualized using $\sqrt{252}$ trading days factor.

4.2 Alpha & Beta Analysis

OLS regression of fund returns against Nifty 100 benchmark using `scipy.stats.linregress`. Alpha represents annualized excess return over benchmark. Beta measures sensitivity to market movements.

4.3 Maximum Drawdown

Maximum drawdown computed as the minimum of $(NAV / \text{running_max} - 1)$ for each fund. This identifies the worst peak-to-trough decline experienced by each fund, along with the exact date range of the drawdown period.

Fund	Sharpe	Alpha	Beta	Max DD
Tata Large Cap	1.39	2.14%	0.98	-28.4%
DSP Midcap	1.48	1.87%	1.12	-31.2%
Nippon India SC	1.49	1.65%	1.21	-35.8%
Canara Robeco	1.50	1.59%	0.95	-26.9%
ICICI Pru Midcap	1.47	1.71%	1.08	-29.7%

5. Advanced Risk Metrics

5.1 Value at Risk (VaR) & CVaR

Historical VaR at 95% confidence level represents the 5th percentile of daily return distribution — the worst daily loss expected 95% of the time. CVaR (Conditional VaR) represents the average loss on the worst 5% of days, providing a more conservative risk measure.

5.2 Rolling 90-Day Sharpe Ratio

Rolling Sharpe analysis reveals how risk-adjusted performance evolves over time. The 2022 market correction caused all funds to show negative rolling Sharpe ratios, recovering strongly through 2023-2024. Small cap funds showed the most volatility in rolling Sharpe.

5.3 Sector HHI Concentration

The Herfindahl-Hirschman Index (HHI) measures portfolio concentration. $HHI = \sum(\text{weight}_i^2)$. A higher HHI indicates more concentrated sector exposure. Axis Midcap Fund shows highest HHI (0.35) — heavily concentrated in Banking and IT sectors.

5.4 Investor Cohort Analysis

Investors grouped by first transaction year show distinct behavior patterns. The 2022 cohort had highest total investment amounts, coinciding with post-COVID market rally. SIP continuity analysis reveals 53 investors (40%) at risk of dropout with average SIP gaps exceeding 35 days.

6. Fund Scorecard & Recommendations

The composite fund scorecard combines multiple metrics into a single 0-100 score using weighted ranking:

Metric	Weight	Rationale
3-Year CAGR Rank	30%	Primary return metric over medium term
Sharpe Ratio Rank	25%	Risk-adjusted return efficiency
Alpha Rank	20%	Excess return over benchmark
Expense Ratio Rank (inverse)	15%	Lower cost = better value
Max Drawdown Rank (inverse)	10%	Capital protection measure

Rank	Fund	Score	CAGR 3yr	Sharpe
1	Tata Large Cap Fund	88.88	85.28%	1.39
2	DSP Midcap Fund	86.25	75.53%	1.48
3	Nippon India Small Cap	82.62	65.72%	1.49
4	Canara Robeco Bluechip	81.25	61.53%	1.50
5	ICICI Pru Midcap Fund	80.75	72.15%	1.47

7. Dashboard Development

A 4-page interactive Power BI dashboard was developed to visualize all analytics findings with drill-through capabilities, slicers, and the Bluestock color theme (#1B3A6B primary, #00B4D8 accent).

Page	Title	Key Visuals
1	Industry Overview	4 KPI Cards, AUM Trend Line, AUM by AMC Bar Chart
2	Fund Performance	Return vs Risk Scatter, Fund Scorecard Table, NAV vs Benchmark
3	Investor Analytics	State Bar Chart, Type Donut, Age Group Bar, Monthly Volume
4	SIP & Market Trends	Dual-axis SIP+Nifty, Category Heatmap, Top 5 Bar Chart

8. Key Findings & Insights

Finding 1: Top Performing Fund

Tata Large Cap Fund emerges as the top performer with a composite score of 88.88/100, driven by exceptional 3-year CAGR of 85.28% and strong alpha generation of 2.14% annualized.

Finding 2: Risk-Adjusted Leaders

Canara Robeco Bluechip achieves the highest Sharpe ratio (1.50) among all 40 funds, indicating superior risk-adjusted returns relative to the 6.5% risk-free rate benchmark.

Finding 3: Small Cap Risk Profile

Small cap funds show VaR exceeding 2% per day and maximum drawdowns of 35%+, confirming their high-risk nature. Only suitable for investors with High risk appetite and 5+ year horizon.

Finding 4: SIP Continuity Challenge

40% of investors with 6+ SIP transactions show average gaps exceeding 35 days, flagged as At-Risk. This represents a significant retention challenge requiring automated reminder systems.

Finding 5: Sector Concentration Risk

Axis Midcap Fund (HHI: 0.35) shows highest sector concentration in Banking and IT, creating correlation risk during sector-specific downturns.

9. Limitations

- **Simulated Data:** NAV data was generated using statistical models rather than actual AMFI data, which may not capture real market microstructure effects.
- **Investor Data:** Investor transaction records are synthetic, limiting the applicability of cohort and SIP continuity findings to real scenarios.
- **Expense Ratios:** Expense ratios are randomly generated within realistic ranges but not sourced from actual fund factsheets.
- **Benchmark Selection:** Only Nifty 50 and Nifty 100 used as benchmarks; category-specific benchmarks (e.g., Nifty Midcap 150) would provide more accurate alpha calculations.
- **Survivorship Bias:** Analysis includes only currently active funds, potentially overstating historical returns by excluding funds that were wound up.

10. Recommendations

For Low Risk Investors

Allocate to Large Cap funds with Sharpe > 1.4 and VaR $< -1.0\%$. Recommended: Tata Large Cap, Canara Robeco Bluechip, HDFC Top 100.

For Moderate Risk Investors

Blend of Large Cap (60%) and Flexi Cap (40%) funds. Focus on funds with positive Alpha and Beta close to 1.0.

For High Risk Investors

Mid and Small Cap funds offer higher returns but with VaR exceeding 2% per day. Recommended: DSP Midcap, Nippon India Small Cap with 5+ year horizon.

For Platform Improvements

Implement automated SIP reminder system for at-risk investors. 40% dropout rate represents significant AUM loss potential.

For Portfolio Construction

Avoid highly concentrated funds ($\text{HHI} > 0.2$). Prefer multi-sector diversified funds to reduce correlation risk during sector downturns.

End of Report

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